THE DERIVED TENSOR PRODUCT OF BOUNDED ABOVE COMPLEXES WITH LOCALLY CONSTANT FINITE TYPE COHOMOLOGY HAS LOCALLY CONSTANT FINITE TYPE COHOMOLOGY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the truncation reduction to bounded complexes, the reduction to complexes of locally constant sheaves of finite type, and the final reduction to finite free modules where the tensor product is computed directly.

Editorial note. The label lemma-locally-constant-tensor-product names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source sites-cohomology.tex –label lemma-locally-constant-tensor-product –start 14355 –end 14376 –local-id 3560 returned RESULT=OK (L3560 free, occupancy FREE, public interval clear). Candidate taken from lane-s4/halfB-priority.json.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Representation of complexes with locally constant cohomology. Same chapter, the lemma showing that a bounded above complex whose cohomology sheaves are locally constant of finite type is represented by a bounded above complex of locally constant sheaves of finite type. Supplies the passage from K and L to complexes of locally constant sheaves of finite type used in the second reduction.
- Vanishing of the cohomology of a truncation product. Derived Categories, the standard truncation statement identifying H^i of a tensor product with H^i of the tensor product of the truncations below i-1. Supplies the first reduction, which lets one assume the complexes bounded.

Source reference. Original reference 093T: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 14247–14247. Prerequisite pointer invoked at proof line(s) 14366; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:sites-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let Λ be a Noetherian ring. Let $K, L \in D^-(\mathcal{C}, \Lambda)$. If the cohomology sheaves of K and L are locally constant sheaves of Λ-modules of finite type, then the cohomology sheaves of $K \otimes^{\mathbf{L}}_{\Lambda} L$ are locally constant sheaves of Λ-modules of finite type.*

Proof. We'll prove this as an application of Lemma 093T. Note that $H^i(K \otimes^{\mathbf{L}}_{\Lambda} L)$ is the same as $H^i(\tau_{\geq i-1}K \otimes^{\mathbf{L}}_{\Lambda} \tau_{\geq i-1}L)$. Thus we may assume K and L are bounded. By Lemma 093T we may assume that K and L are represented by complexes of locally constant sheaves of Λ-modules of finite type. Then we can replace these complexes by bounded above complexes of finite free Λ-modules. In this case the result is clear. □